Velocity

Speed is the distance and object travels in a set amount of time. Velocity and **speed** are very similar ideas, but velocity is a vector, and speed is not. Suppose we knew that someone was driving at thirty-five kilometres an hour (35 km/hr), but the direction wasn't given. How would you draw an arrow to represent a vector? You can't know how to draw the vector if you only have one value (either amount or direction). In this example, you were never told about the direction. Physicists would say that the speed is thirty-five kilometres an hour (35 km/hr), but the velocity is unknown. On the other hand, if you're moving at 35 km/hr in a northern direction, then you would have an arrow pointing north with a length of 35. Physicists would say that the velocity is 35 km/hr north.

Velocity is the rate of motion in a specific **direction**. I'm going that-a-way at 30 kilometers per hour. My velocity is 30 kilometers per hour that-a-way. Average speed is described as a measure of distance divided by time. Velocity can be constant, or it can change (acceleration). **Speed with a direction is velocity**.

Remember vectors? You will use a lot of vectors when you work with velocity. Our real world example of navigation on the ocean used velocity for every vector. Velocity is a vector measurement because it has an amount and a direction. Speed is only an amount (a scalar). Speed doesn't tell the whole story to a physicist. Think of it another way. If I tell you I'm driving north and ask you how long until we get to the city. You can't know the answer since you don't know my speed. You need both values.

Acceleration is the rate at which an object changes its speed or velocity . Suppose you are riding your bike. You start out going very slowly, hardly pedalling at all. Now you begin to pedal as hard as you can, to speed up - you are accelerating. Now that you are going at a normal speed, you stop pedalling so hard, and just pedal normally. You're still going, but you're not getting any faster, just going along at your normal speed. You're not accelerating anymore.

If you stop pedalling now, friction will work on your bike tires (and you'll have friction from the air, too), and you'll soon start to go slower. That's negative acceleration, or deceleration. You're still going, but you are slowing down.

One important cause of acceleration is gravity. Suppose you dive off a cliff or a high diving board. You will start off falling slowly, but as gravity pulls on you, you will speed up (accelerate) until you are going very fast.

The acceleration of Earth's gravity will speed you up at about 9.8 meters per second per second (9.8 m/s^2, or 9.8 meters per second squared). That means that for every second you fall, you'll be going 9.8 meters/second faster.

Standing on the cliff before you jump, you're going zero meters/second. One second after you jump, you'll be going 9.8 meters/second. Two seconds after you jump, you'll be going 19.6 meters/second. Three seconds after you jump, you'll be going 29.4 meters/second.

Key words

Word	Meaning
Average speed	The distance an object travels in a set amount of _____.
Velocity	
Acceleration	

What causes acceleration on a bicycle?

What causes deceleration (negative acceleration) on a slowing bicycle?

What causes the acceleration of a falling ball?

What causes deceleration (negative acceleration) on a slowing ball?

V Z T R E F C B I B T N F C S
D E C E L E R A T I O N R I T
S Q L L P V Z D O I B B I T I
J C O O O A I H T P Y B C E C
T Y I G C S T A N T G X T N K
D X J S T I R F I R F A I I E
V P M A Y E T V M E N A O K R
X M N J L H A Y K O C Z N B I
X C E E R R P V E C T O R W F
E N C W G E M I T U D I V Y R
I C F N H Q Z F L C H O O S T

A Y J P Y T I T N A U Q P N C
C Q Y V S O Z P E D O E G E J
L D C D T L R I U Y E W B X X
B H Z M J T T W H D X S F T F

ACCELERATION	DECELERATION	DISTANCE
FRICTION	GRAVITY	KINETIC
MOTION	PHYSICS	QUANTITY
SPEED	TAPE	TICKER
TIME	VECTOR	VELOCITY